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Biochemistry & Biophysics

Angiotensin II Enhances Activity and Flexibility in Murine $CD8^+$ T Cell Immunometabolism

Audrey Bai (1,2,4), Quinn Hulsey (1), Raghad AlMotairy (1), Allison Norlander (1,2,3)

[1]Department of Anatomy, Cell Biology, and Physiology, IU School of Medicine, Indianapolis, IN, USA
[2]Melvin and Bren Simon Comprehensive Cancer Center, IU School of Medicine, Indianapolis, IN, USA
[3]Krannert Cardiovascular Research Center, IU School of Medicine, Indianapolis, IN, USA
[4]Department of Biology, Amherst College, Amherst, MA, USA

Purpose. Cytotoxic ($CD8^+$) T cells play a crucial role in adaptive immunity against intracellular pathogens and tumors by killing infected and mutated cells. Persistent antigen exposure, however, can induce an "exhausted" $CD8^+$ phenotype characterized by inhibitory receptors, impaired cytokine production, and poor proliferative potential and survival. As such, exhaustion hinders an effective immune response against cancerous cells, leading to worse disease prognoses and less effective immunotherapies. In this study, we examined $CD8^+$ exhaustion in response to elevated angiotensin II (Ang II) levels, a phenotype implicated in many malignancies. We expected the CD8s of mice with elevated levels of Ang II to exhibit increased exhaustion with a less active and robust metabolic profile, due to the bidirectional relationship between hypertension and chronic inflammation.

Methods. We implanted subcutaneous osmotic pumps containing Ang II or vehicle into 21-day-old male C57BL/6 wild-type mice, incubating for 28 days. We then ran SCENITH analysis of CD8s isolated from harvested mouse spleens. SCENITH (Single Cell ENergetIc metabolism by profilIng Translation inHibition) incorporates flow cytometry to reveal various cell populations' metabolic profiles.

Results. The mice of the Ang II treatment group (which were shown to be hypertensive from characteristic fibrosis evident in renal and aortic histology slides) had bulk CD8s that, despite elevated levels of exhaustion markers, exhibited a decreased mitochondrial dependence (MD), a increased glycolytic capacity (GC), and an increased fatty acid oxidation/amino acid oxidation capacity (FAO/AAO cap). When we compared just the exhausted CD8s of the two treatment groups, once again, the hypertensive CD8s exhibited a decreased glycolytic dependence (GD), a decreased MD, an increased GC, and an increased FAO/AAO cap. This was the trend for both examined exhausted CD8 populations.

Conclusion. Even though Ang II is associated with increased CD8 exhaustion, our results demonstrate its role in enhancing metabolic activity and flexibility in $CD8^+$ T cells, especially in the exhausted subsets. This reveals the need for further elucidation and unraveling of the various effects of Ang II, including its role exacerbating exhaustion, acting as a co-stimulatory molecule in CD8 activation, promoting tumor growth and metastasis, and as a mediator of hypertension and inflammation, etc.

Biochemical Characterization of Human LIG1 Syndrome Variants as Possible Players in Cancer Expression

Kellen Garris (1), David Beier (2), & Patrick O'Brien (2)

[1]Department of Neuroscience, Amherst College, Amherst, MA, 0100
[2]University of Michigan Department of Biological Chemistry, Ann Arbor, MI 48109 USA

Purpose. Human DNA ligase 1 (LIG1) uses ATP and magnesium in the final steps of nucleotide excision repair, long-patch base excision repair, and normal DNA replication. Mutation of this enzyme can lead to human LIG1 syndrome, characterized by immunodeficiency and developmental delays. There is growing evidence for the involvement of R305Q, R771G, and R771W LIG1 syndrome mutations in genome instability and non-specific patient cancers. Determining whether rare LIG1 mutants increase cell mutagenesis by shifting enzyme fidelity would underscore LIG1's involvement in cancer formation and highlight a relatively new therapeutic target for cancer prevention.

Methods. Steady-state kinetics competition assays were run before using urea polyacrylamide gel electrophoresis (PAGE) for product separation and quantification. Assays were run at 37℃ and quenched in FLS. An analysis of initial turnover rates allowed me to look at how well these different LIG1 mutants discriminated against sealing oxidized (damaged) DNA single-stranded breaks and preferentially closed the backbone of healthy canonical DNA.

Results. All mutants were able to better discriminate against G:A mismatches compared to WT LIG1. As cofactor increased, fidelity decreased for all LIG1 species except R771G.

Conclusion. The LIG1 variants tested all have higher fidelity than WT using a 3' 8oxoG:A pair. These mutants appear unlikely to increase cancer risk than WT LIG1. While R771W and R305Q are sensitive to increasing free [Mg2+], R771G appears resistant to increasing free cofactor. This outcome favors a LIG1 syndrome model in which longer-living single-stranded breaks are a source of mutations.

Impact of Carbohydrate Sources On Prophage Induction In Phocaeicola vulgatus

Theint Theint Thu(1,2), Saria McKeithen-Mead(1), Jiawei Sun(1), Kerwyn C. Huang(1)

[1]Department of Bioengineering, Stanford University, Stanford, CA 94305
[2]Amherst College, Amherst, MA 01002

Purpose. In a previous murine model of a stable synthetic gut community, a mobile genetic element (MGE) identified as a prophage transferred from the invading microbe P. vulgatus to a dominant community member. This transfer coincided with diet-dependent prophage induction, with increased copy number variation (CNV) observed in mice fed a standard diet but not a high-sugar diet. Here, we investigated how carbon source and concentration regulate prophage induction and lysis dynamics of P. vulgatus in vitro.

Methods. Follow-up in vitro experiments tested P. vulgatus growth in defined minimal media with various carbon sources such as D-glucose, L-arabinose, or rhamnogalacturonan-I (RG1) at concentrations ranging from 0.25–5 mg/mL.

Results. Growth curves showed that at 5 mg/mL, lysis did not occur despite entry into the stationary phase, suggesting alternative regulatory pathways limit activation under saturating carbon conditions. Lysis was most pronounced at 2.5 mg/mL, and induction occurred even at 0.25 mg/mL. Phase contrast and LIVE/DEAD microscopy confirmed reduced viability post-lysis. qPCR targeting phage excision (attP) revealed prophage induction in a small subset of cells, with CNV increases preceding lysis.

Conclusion. These findings support that prophage induction in P. vulgatus is carbon source and concentration-dependent, and regulated by mechanisms beyond simple growth phase entry. Future work will involve expanding qPCR to measure phage particles for better understanding of what goes on at each time point in the growth assay, curing P. vulgatus of the prophage to assess fitness effects, and performing colony-forming unit assays to complement OD_{600} measurements.